

FINAL EXPLANATION: BIOLOGY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY | GRADE 10 | 2.1 PLANT NUTRITION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the sub-strand. Think back to Amina's sukuma wiki in her Nairobi kitchen garden: it wilts at midday even when the soil is moist, recovers overnight, and turns pale and grows slowly when moved into the shade of a mango tree. Use evidence from all your lessons (leaf structure, what plants need for photosynthesis, the starch test, the light-dependent and light-independent reactions, limiting factors, and mineral nutrition) to answer each question in the space below it. Write in full sentences, use correct scientific terms, and link every answer back to the sukuma wiki. Where a question asks for a claim, give your evidence and reasoning. Where it asks for a diagram or table, draw it neatly.

Part 1: How do plants get their food?

Amina's sukuma wiki and a mushroom growing nearby get their food in different ways. (a) Name the type of nutrition each organism uses and explain the difference between them. (b) State what a green plant needs for photosynthesis: which of these are materials used up as reactants and which are not? For each, say how it reaches the leaf or is used there. (c) Write the word equation and the balanced chemical equation for photosynthesis.

(a) The sukuma wiki is an autotroph: it makes its own food from simple inorganic raw materials. The mushroom is a heterotroph: it cannot make its own food and depends on ready-made organic food from other sources. (b) The raw materials (reactants) are carbon dioxide and water. Carbon dioxide enters the leaf through the stomata and diffuses through the air spaces of the spongy mesophyll. Water is absorbed by the roots and carried up the xylem to the mesophyll cells. Two further things are also needed but are not used up as reactants: light energy, which comes from the sun, and chlorophyll, the green pigment in the chloroplasts of palisade cells that absorbs the light energy. Chlorophyll is not consumed in the reaction; it is essential as a light-absorbing pigment. (c) Word equation: carbon dioxide + water, in the presence of light energy and chlorophyll, gives glucose + oxygen. Chemical equation: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (with light energy and chlorophyll).

Part 2: Leaf structure and the wilting puzzle

(a) Choose THREE leaf structures you saw in the cross-section of the mango or bean leaf. For each, describe how its structure suits its function. (b) Use what you know

(a) Any three of: Palisade mesophyll cells are tightly packed and contain many chloroplasts, so they capture a lot of light for photosynthesis. The spongy mesophyll has air spaces, so CO_2 and O_2 can diffuse easily through the leaf. Stomata with guard cells can open and close, so they regulate gas exchange and water vapour loss. The transparent cuticle and epidermis let light through while reducing water loss. The xylem carries water and minerals up to the leaf, and the phloem carries sugars away. (b) In the strong midday sun the stomata are open to take in CO_2 for photosynthesis, and open stomata let water vapour escape from the leaf by transpiration. When water is lost through the stomata faster than the xylem can bring water up from the roots, the leaf

about stomata, transpiration, xylem and turgor to explain why Amina's sukuma wiki wilts at midday even though the soil is moist. (c) Explain why the plant recovers overnight.	cells lose water and lose turgor pressure, so the leaves droop. This can happen even when the soil is moist, because the problem is the rate of water loss compared with the rate of supply, not the amount of water in the soil. Wilting can continue even if the guard cells begin to close the stomata partly, because the leaf has already lost more water than the xylem has replaced. (c) Overnight there is no strong sunlight, the stomata close and transpiration slows right down. The xylem can then catch up and refill the leaf cells with water. Turgor is restored and the leaves become upright again by morning. The wilting is reversible because the cells were not permanently damaged.
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Part 3: Evidence from the starch test and the two stages of photosynthesis

(a) In the starch test on a variegated or partly covered leaf, describe the iodine result in each of the three regions: white region, covered green region, and uncovered green region. What does each result tell you? (b) Describe what happens in the light-dependent reactions and in the light-independent reactions (Calvin cycle), including where each takes place and how the two stages are linked. (c) Use both stages to build a cause-and-effect chain that explains why the sukuma wiki in the shade of the mango tree grows slowly and turns pale.	(a) White region (no chlorophyll): it stayed brown-orange, so no starch was made, showing chlorophyll is needed. Covered green region (chlorophyll but no light): it stayed brown-orange, so no starch was made, showing light is needed. Uncovered green region (chlorophyll and light): it turned blue-black, so starch was present. Both chlorophyll and light must be present together for photosynthesis. (b) The light-dependent reactions take place in the thylakoid membranes of the chloroplast. Chlorophyll in Photosystems II and I absorbs light and excites electrons, which pass along the electron transport chain. Water is split (photolysis: $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$), releasing oxygen and supplying electrons. ATP is made by photophosphorylation, and NADP ⁺ is reduced to NADPH. The light-independent reactions (Calvin cycle) take place in the stroma. CO ₂ is fixed onto RuBP by the enzyme RuBisCO to form 3-PGA. The 3-PGA is reduced to G3P using the ATP and NADPH from the light stage. Some G3P is used to make glucose and some regenerates RuBP. The ATP and NADPH link the two stages. (c) Less light under the mango tree means fewer electrons are excited in the thylakoids, so less ATP and NADPH are produced. The Calvin cycle then cannot reduce 3-PGA to G3P efficiently, so less glucose is made. With less glucose for energy and building materials, cell division and growth are slow. The plant also has less energy and fewer building materials available to make and maintain chlorophyll, so the leaves gradually lose their green pigment and become pale yellowish-green over several weeks.
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Part 4: Limiting factors and Kenyan farming

(a) In the Lesson 7 investigation you counted oxygen bubbles at lamp distances of 5, 10, 20, 30 and 40 cm. Describe the shape of the graph of rate of photosynthesis against light intensity, and explain why the line flattens at high light intensity. What is a limiting factor? (b) Name the other factors you studied that affect the rate of photosynthesis, and describe how temperature	(a) The rate of photosynthesis (bubbles per minute) increases as light intensity increases (a lamp closer to the plant means higher intensity), but only up to a point. Then the curve flattens, because another factor, such as CO ₂ concentration or temperature, has become the limiting factor and extra light cannot increase the rate further. A limiting factor is the factor present at the lowest level, which controls the rate of the process. (b) The other factors are CO ₂ concentration and temperature. As temperature rises the rate increases because enzymes work faster, until an optimum is reached. Above about 40 °C the enzymes denature and the rate falls, giving a bell-shaped curve. (c) In well-lit, warm Naivasha greenhouses, light and temperature are already favourable, so CO ₂ is the limiting factor. Adding CO ₂ raises the rate of photosynthesis and the tomato yield. In Kericho, temperature, not CO ₂ , is the limiting factor. Adding CO ₂ will not significantly increase the rate, because the factor in shortest supply still controls the rate.
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changes the rate. (c) Greenhouse tomato farmers in Naivasha add CO ₂ , but adding CO ₂ to the tea farms in Kericho would not help much. Explain why using the idea of limiting factors.	
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Part 5: Minerals and the complete explanation for Amina	
(a) Name two mineral nutrients taken up from the soil, say how roots absorb them, and describe what each is needed for and what deficiency symptom you would expect. (b) Write a complete explanation answering the driving question: why does the sukuma wiki wilt at midday even when the soil is wet, and what does a plant actually need to stay green, grow, and make its own food? Your answer must explain both the wilting and the pallor in the shade. (c) Give one practical piece of advice, based on your explanation, that a Kenyan smallholder farmer could use to improve sukuma wiki yield.	(a) Nitrogen is needed to make proteins and chlorophyll; a deficiency causes chlorosis (pale yellow leaves) and stunted growth. Magnesium is needed to make chlorophyll; a deficiency causes yellowing of leaves. (Phosphorus, potassium and others are also acceptable with correct functions and symptoms.) Minerals dissolved in soil water are absorbed through the root hair cells by active transport. (b) Wilting: in the strong midday sun the stomata are open for CO₂ to enter for photosynthesis, and water vapour is lost by transpiration. When water leaves through the stomata faster than the xylem can deliver it from the roots, the leaf cells lose turgor and the plant wilts, even though the soil is moist. Overnight the stomata close, transpiration slows, the xylem catches up, turgor is restored, and the plant recovers. Pallor in the shade: less light under the mango tree reduces the light-dependent reactions, so less ATP and NADPH are made and the Calvin cycle produces less glucose. The plant has less food for growth, so it grows slowly, and it has less energy and fewer building materials to make and maintain chlorophyll, so the leaves turn pale. Low light is the limiting factor. Minerals matter too: nitrogen and magnesium are needed to build chlorophyll, so the soil must supply them for the leaves to stay green. To stay green, grow and make its own food, the plant needs light, chlorophyll, water and CO₂, plus minerals such as nitrogen and magnesium from the soil, and when any one is limiting, photosynthesis slows. (c) For example: water the sukuma wiki in the early morning or evening and not at midday, so that water supply keeps up with loss; plant it in a place with enough sunlight rather than deep shade; and apply a fertiliser or manure containing nitrogen so the leaves can make chlorophyll and stay green.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of photosynthesis (reactants, light and chlorophyll, equations, two stages)	Correctly distinguishes the reactants from light energy and chlorophyll, gives both equations, and accurately describes where each stage occurs, the key steps and how ATP and NADPH link the stages.	States most requirements and equations correctly and describes both stages with minor omissions or errors.	Names some requirements or an equation but confuses the stages or leaves out key steps.
Explaining wilting using leaf structure and water movement	Clearly links stomata, transpiration, xylem supply rate and loss of turgor, explains why wet soil does not prevent	Explains wilting through water loss and turgor but is less clear about rate of loss	Says the plant lacks water or is hot without linking stomata, transpiration and

	wilting, and explains overnight recovery.	versus supply or about recovery.	turgor.
Use of evidence and limiting factors	Accurately uses the starch test, the light-intensity graph shape, the temperature curve and the definition of a limiting factor to justify claims, and correctly applies them to the Naivasha and Kericho examples.	Uses some evidence and defines a limiting factor, with partial application to the farm examples.	States facts without evidence, or the limiting factor idea is missing or incorrect.
Integration of minerals and the complete causal chain for the pallor	Builds a clear, sequenced chain from low light to less ATP and NADPH to less glucose to slow growth and less chlorophyll maintained, and explains the role of nitrogen and magnesium in chlorophyll.	Gives a mostly complete chain but one step or the link to chlorophyll or minerals is missing or unclear.	Gives a brief statement that shade means less food, without a causal chain.
Communication and Kenyan application	Uses precise scientific vocabulary, organises answers clearly, and gives a sensible, evidence-based farming recommendation.	Mostly clear writing with some correct terms and a reasonable but general recommendation.	Writing is unclear or uses few scientific terms, and the recommendation is missing or not linked to the evidence.

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Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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